Python Analyzers

Set - C

Branch: _____ **Date:** 24-01-2026

Total Marks: 50 **Time:** 2.5 Hours

Sr. No.	GR ID	Name	Signature
1			

Rules & Guidelines

- Participants must bring their own laptop
- Use of the internet is strictly restricted during the competition (apart from the final submission)
- Any form of plagiarism or code sharing will lead to disqualification
- Participants must strictly follow the instructions given during the event
- Late submissions will not be accepted
- Maintain discipline during the competition
- The decision of the faculty/judges will be final

Submission Details

- **File format:** .ipynb file
- **File name format:** Name_GRID_PythonAnalyzers.ipynb
- **Submission method:** Submit the **.ipynb file** and the given **dataset (.csv/.xlsx)** and upload it to the GitHub repository. **Submit that repo link** in the **Google form** for submission.
- Submit here: **LINK**

🛠️ Mandatory Technologies

Students **must use only**:

- numpy
- pandas
- matplotlib
- seaborn

❌ Any other ML / AI libraries are **not allowed**

📂 Dataset Overview (Gigantic Real-World Dataset)

Dataset Name:
[Healthcare_Hospital_Operations_Gigantic_Dataset.csv]

Dataset Size:

- ~65,500+ Rows
- 10+ Attributes

Dataset Description

This dataset represents **hospital operations and patient management data across multiple Indian regions**, suitable for **healthcare analytics and operational performance analysis**.

Columns Explanation

Column Name	Description
Patient_ID	Unique patient identifier
Admission_Date	Date of hospital admission
Hospital_Region	Region of hospital
City	City of hospital
Department	Medical department
Patient_Type	Inpatient or Outpatient

Length_of_Stay_Days	Duration of stay
Treatment_Cost	Total treatment cost
Insurance_Type	Insurance category
Discharge_Status	Final patient outcome

🎯 Project Objective

Analyze the provided **hospital operations dataset** to:

- Clean and prepare real-world healthcare data
- Perform exploratory and operational analysis
- Identify efficiency bottlenecks and cost drivers
- Visualize healthcare trends and patterns
- Present meaningful hospital performance insights

📝 Mandatory Tasks (Exam Tasks for Students)

🔹 Task 1: Notebook Structure & Data Loading

- Proper title & introduction
- Import all required libraries
- Load the dataset correctly
- Display:
  - Dataset shape
  - First & last 5 rows
- Proper Markdown usage

🔹 Task 2: Data Cleaning & EDA

Students must:

- Identify and handle missing values
- Validate data types
- Detect and handle duplicate records
- Generate:
    - Summary statistics
    - Department-wise and region-wise distributions
- Comment on initial observations

🔹 Task 3: Healthcare / Operational Data Analysis

Perform **minimum 5 analytical tasks**, such as:

- Department-wise patient volume
- Average treatment cost per department
- Region-wise hospital load
- Patient type vs length of stay
- Insurance type vs treatment cost analysis

🔹 Task 4: Data Visualization

Create **minimum 5 meaningful visualizations**, such as:

- Bar chart (Department vs Patient Count)
- Line chart (Admissions trend over time)
- Histogram (Treatment cost distribution)
- Pie Plot
- Scatter Plot
- Subplots (Combine multiple charts)

Each chart must include:

- Title
- Axis labels
- Insight explanation

🔹 Task 5: Final Insights & Task Completion

- Summarize key findings
- Mention healthcare insights

- Provide hospital improvement recommendations

📊 Evaluation & Marking Scheme (50 Marks)

Criteria	Marks
Notebook Structure & Data Loading	10
Data Cleaning & EDA	10
Data Analysis	10
Data Visualization	10
% Task Completion & Overall Quality	10
TOTAL	**50**

⚠️ Mandatory Rules

- Only .ipynb file accepted
- No plagiarism or copied notebooks
- Code must be readable & logical
- Insights are more important than quantity
- The Internet is allowed only for the final submission

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources.

Good luck with your project work!

Python Analyzers
Data Science & AI/ML Department

BRING ON YOUR CODING ATTITUDE